

PROJECT OVERVIEW

Advanced AI-Integrated Automation of CRMG and STS Cranes for Optimized Port Operations

Developed as a B.Tech Capstone Project

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1. Introduction

Modern seaports are critical infrastructure assets that facilitate global trade and logistics. As cargo volumes continue to increase, container terminals face growing pressure to improve operational efficiency, safety, and throughput.

This project was developed after studying the operational workflow of container handling systems used at Adani Vizhinjam Port, one of India's emerging transshipment hubs. Ship-to-Shore (STS) and Cantilever Rail-Mounted Gantry (CRMG) cranes play a central role in moving containers between vessels, internal transit vehicles, and storage yards.

Inspired by the challenges associated with large-scale container handling, this project explores the application of Artificial Intelligence, Computer Vision, IoT, and embedded control systems to automate crane operations. The proposed system demonstrates how real-time visual monitoring and intelligent control can be used to improve container tracking, reduce manual intervention, enhance operational safety, and optimize cargo movement within modern container terminals.

The project was implemented as a scaled prototype and serves as a proof-of-concept for future intelligent port automation systems.

2. Problem Statement

Container terminals operate in dynamic environments where rapid and accurate cargo movement is critical.

Traditional crane control systems face several challenges:

- Dependence on manual supervision
- Delays caused by operational inefficiencies
- Limited adaptability to changing container positions
- Increased risk of human error
- Higher vessel turnaround times
- Reduced operational flexibility

There is a growing need for intelligent automation systems capable of making real-time decisions and coordinating crane operations with minimal human intervention.

3. Objectives

The primary objectives of this project are:

- Automate container handling operations.
- Improve loading and unloading efficiency.
- Reduce operational delays.
- Enhance safety by minimizing human intervention.
- Enable real-time container detection and tracking.
- Improve crane positioning accuracy.
- Develop a scalable framework applicable to both STS and CRMG cranes.
- Demonstrate practical integration of AI, IoT, and embedded control systems.

4. Methodology

The project was executed through a phased development approach.

Phase 1 – Manual Prototype Development

- Development of a scaled crane prototype.
- Study of crane movement mechanisms.
- Evaluation of operational workflows.
- Manual control using joystick-based interfaces.

Phase 2 – Semi-Automated Operation

- Integration of computer vision systems.
- Real-time container identification.
- Assisted decision-making for crane positioning.
- Improved motor control architecture.

Phase 3 – Fully Automated Operation

- Real-time visual monitoring.
- Autonomous container detection and tracking.
- Automated gantry, trolley, hoist, and spreader movements.
- Integrated communication between software and hardware subsystems.

5. System Architecture

The overall system consists of six major layers:

1. Visual Data Acquisition
2. Image Processing and Decision Making
3. Control Logic
4. Microcontroller Interface
5. Motor Driver System
6. Crane Motion Actuation

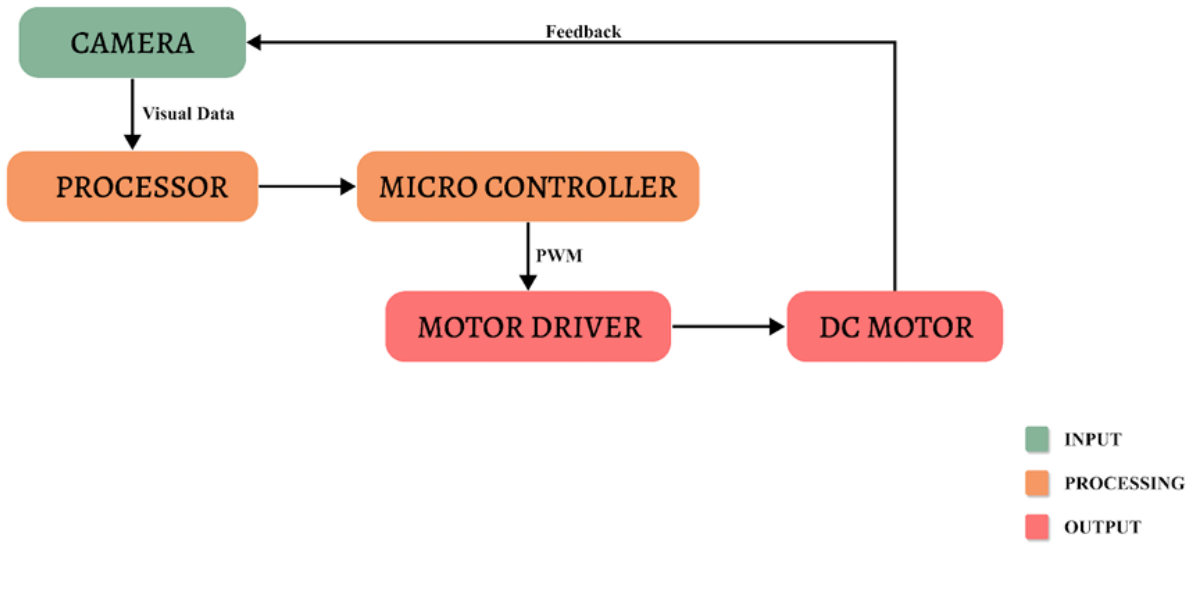


Fig 1: Phase 1 System Architecture

This architecture enables real-time monitoring and automated crane control.

6. Hardware Overview

The prototype incorporates industrial-style control architecture using the following hardware components:

Processing Units

- Intel i5-based Processing System
- Arduino Mega 2560

Vision System

- High-Resolution Cameras

Motor Control

- Custom H-Bridge Driver Circuits
- BTS7970-Based Motor Drivers

Actuation

- Johnson DC Motors

- Geared Motors
- Servo-Based Spreader Mechanism

Power System

- Regulated Power Supply
- Voltage Regulation Circuits

Communication

- Serial Communication Interface
- Embedded Control Architecture

7. Prototype Development

The project progressed from a manually operated prototype to a fully automated crane model.

Major prototype features include:

- Gantry movement automation
- Trolley travel automation
- Hoist movement automation
- Spreader control
- Real-time container tracking
- Automated loading and unloading operations

The prototype successfully demonstrated autonomous crane behavior within a scaled testing environment.



Fig 2: STS Crane Prototype

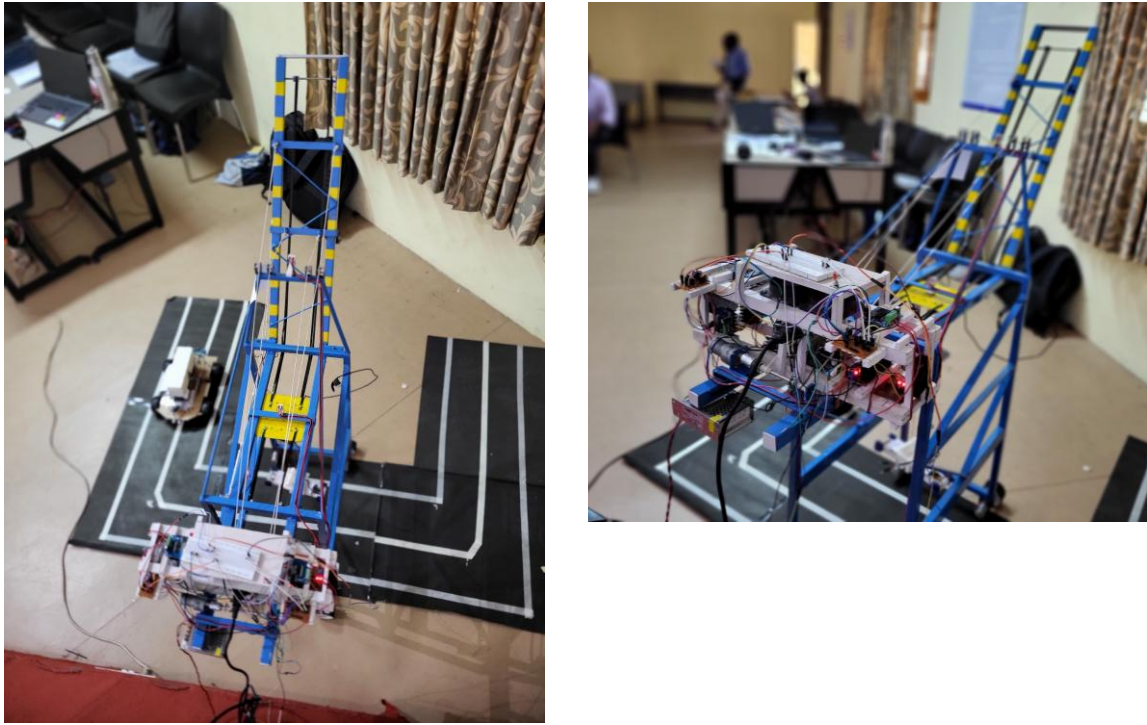


Fig 3: Hardware Control System

8. Results

The developed system successfully demonstrated:

- Real-time container tracking
- Accurate crane positioning
- Autonomous crane movement
- Reduced operator dependency
- Improved operational consistency
- Faster cargo handling workflows
- Enhanced safety through reduced manual intervention

The prototype validated the feasibility of AI-assisted crane automation for modern container terminals and demonstrated reliable operation throughout testing.

9. Future Scope

Several opportunities exist for future expansion:

- Deployment in full-scale industrial environments.
- Integration with cloud-based monitoring systems.
- Predictive maintenance frameworks.
- Fleet-wide crane coordination.
- Advanced scheduling and logistics optimization.
- Digital twin integration.
- AI-driven operational analytics.
- Autonomous yard management systems.

These enhancements can further improve efficiency, scalability, and sustainability in next-generation smart ports.

Conclusion

This project demonstrates the practical application of Artificial Intelligence, Computer Vision, IoT, and embedded control systems in port automation. Through the development of an automated crane control framework, the project showcases a scalable approach to improving efficiency, safety, and productivity in container terminal operations.

The successful prototype establishes a strong foundation for future deployment of intelligent automation technologies in modern seaport infrastructure.